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Web Platform Threats: Automated Detection of Web Security Issues With WPT

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Client-side security mechanisms implemented by Web browsers, such as cookie security attributes and the Mixed Content policy, are of paramount importance to protect Web applications.

Unfortunately, the design and implementation of such mechanisms are complicated and error-prone, potentially exposing Web applications to security vulnerabilities. In this paper, we present a practical framework to formally and automatically detect security flaws in client-side security mechanisms. In particular, we leverage Web Platform Tests (WPT), a popular cross-browser test suite, to automatically collect browser execution traces and match them against Web invariants, i.e., intended security properties of Web mechanisms expressed in first-order logic. We demonstrate the effectiveness of our approach by validating 9 invariants against the WPT test suite, discovering violations with clear security implications in 104 tests for Firefox, Chromium and Safari. We disclosed the root causes of these violations to browser vendors and standard bodies, which resulted in 8 individual reports and one CVE on Safari.

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